

Key Technical Concepts & Tools

- **Developer & DevOps Tools** :

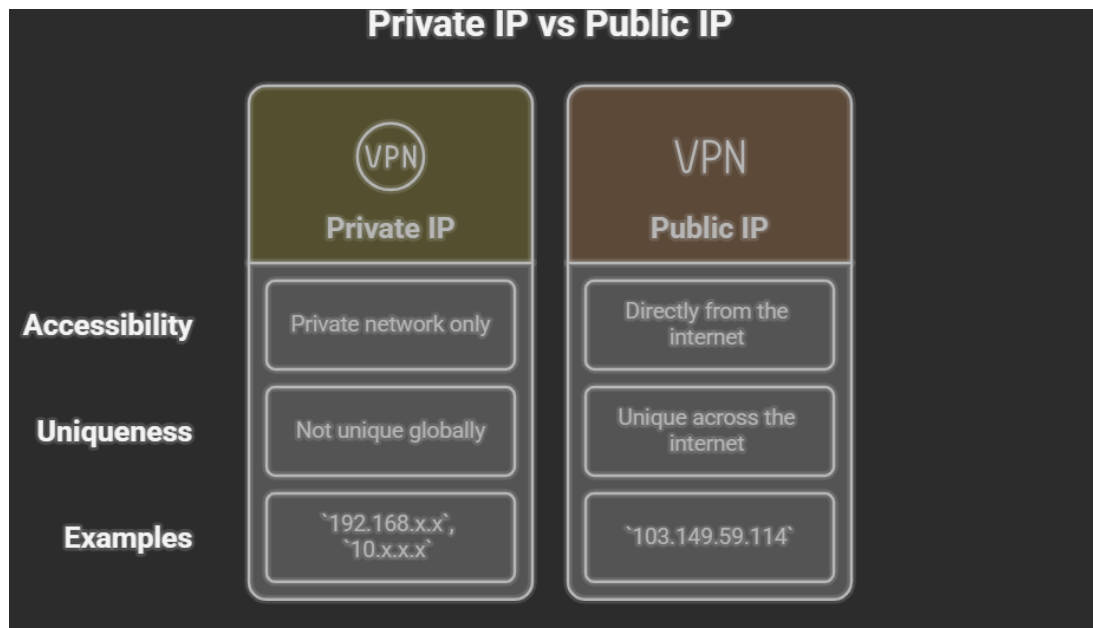
- **Visual Studio:** (VS) - An Integrated Development Environment (IDE) for writing code.
- **Putty:** An SSH and Telnet client for connecting to remote servers.
- **Terraform:** Infrastructure as Code (IaC) tool for building, changing, and versioning infrastructure efficiently.
- **AWS CLI v2:** Amazon Web Services Command Line Interface - used to interact with AWS services from the command line.

- **AMI (Amazon Machine Image)** :

- **Definition:** A template containing a software configuration (operating system, application server, and applications).
- **Analogy:** Like a pre-built snapshot or a mold. "We and our surroundings" captured at a point in time.
- **Use Case:** Used to launch EC2 instances (virtual servers) on AWS. You can create AMIs of your own configured servers (e.g., RedHat 9 + Nginx + apps).

- **Public IP vs. Private IP** :

- **Private IP:** Used within a private network (like your home network or a corporate network). Not accessible directly from the internet. Examples: 192.168.x.x, 10.x.x.x, 172.16.x.x to 172.31.x.x. (e.g., 192.168.1.102)
- **Public IP:** Assigned to a device that is directly accessible from the internet. Unique across the entire internet. (e.g., 103.149.59.114)
- **Diagram Idea:**  (Internet)  **Public IP** (Gateway/Router/Server exposed to internet)  **Private IP** (Machines inside the network)



- **Linux Server & Nginx** 🐧🌐⚙️ :

- **Linux:** An operating system (like Windows or macOS) often used for servers. It runs on the physical (or virtual) machine.
- **Nginx:** A popular web server and reverse proxy. It's a *virtual server* software application that *runs on* the Linux operating system. It handles incoming web requests (HTTP).
- **Installation Commands:**
 - `yum install nginx -y` (Older command for RHEL/CentOS/Amazon Linux)
 - `dnf install nginx -y` (Newer, advanced command for RHEL 8+, Fedora; supports modules)
 - `apt-get install nginx -y` (Command for Debian/Ubuntu)
- **Configuration File:** `/etc/nginx/nginx.conf` - contains Nginx settings (which ports to listen on, where website files are, etc.). **Need to restart the Nginx service** after changing this file (`systemctl restart nginx` or `service nginx restart`).
- **Default Website Root:** `root /usr/share/nginx/html;` - This line in the config typically specifies the directory where Nginx looks for the website files (like `index.html`).
- **Accessing Nginx:** `http://<Public_IP>:80` - Accessing the web server via its public IP address on the default HTTP port (80). (e.g., `http://44.201.38.173:80`)

- **Domain Names & Registrars** 🏷️ ➡️ 1234 :
 - **Domain Name:** A human-readable name that maps to an IP address (e.g., facebook.com maps to 143.234.53.98).
 - **Domain Registrar:** Companies where you register domain names (e.g., GoDaddy, Hostinger). You pay an annual fee to keep the domain name.
 - **Cost:** Often cheap for the first year (hostinger mentioned as cheaper initially), but renewal costs can be higher.
- **Basic Computer Representation** 01 🔴 :
 - Computers fundamentally understand **binary** (0s and 1s).
 - These binary numbers are often converted to **decimal** (base 10) for easier human understanding.
 - IP addresses (like 143.234.53.98) are numerical addresses for devices on a network, acting like phone numbers for computers.

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